# TOTAL QUALITY MANAGEMENT (TQM) IN EDUCATIONAL INSTITUTIONS

Case Study: Eradicating Paper Dependency & Elevating Institutional Quality via a Digital ERP Portal

**Academic Project Report**

School / College Management System

Prepared from the supplied TQM_Project_Presentation_Guide.md

**Note: The quantitative figures in this report reproduce the figures supplied in the project guide. They should be treated as project baseline/benchmark figures and independently validated before being presented as externally verified empirical findings.**

# Executive Summary

Educational institutions often depend on paper registers, physical fee receipts, printed report cards, manual leave applications, and disconnected administrative processes. These practices can create delays, duplicated work, record loss, reconciliation disputes, and poor communication with parents. This project proposes a unified web-based School Management System guided by Total Quality Management (TQM).

The proposed portal covers six operational domains: attendance, fees, academics, timetable, student promotion, and parent–teacher meeting (PTM) leave workflows. TQM concepts—including customer focus, Deming's chain reaction, Juran's Quality Trilogy, Crosby's Zero Defects, the Kano model, SERVQUAL, the seven QC tools, 5S, process capability, and Cost of Quality—are mapped to practical digital controls.

**Project objective: replace avoidable paper-based workflows with traceable, validated, accessible digital processes while establishing continuous improvement through measurement and stakeholder feedback.**

# 1. Problem Statement and Objectives

**Problem statement.** Many rural and tier-2/3 urban schools continue to depend heavily on manual paperwork for routine operations. This can consume staff time, make information harder to retrieve, and create avoidable errors in fees, attendance, marks, and parent communication.

**Figures stated in the supplied guide:** over 78% of target schools rely on manual paper processes; 45+ staff hours per week are spent on clerical data entry; a 14.2% historical record-loss rate; 23% fee reconciliation dispute rate; and 68% parent defection/dissatisfaction attributed to administrative friction. These figures are project-guide figures and are not independently verified in this generated report.

**Objectives**

- Digitize high-frequency administrative workflows.
- Reduce manual data entry and repeated handling of records.
- Improve accuracy, traceability, access control, and data availability.
- Improve service turnaround time for parents, students, teachers, and administrators.
- Apply measurable TQM methods to detect defects and drive continuous improvement.

# 2. Proposed Digital ERP Portal

The proposed system brings six operational domains into a common portal. A role-based interface can support teachers, accounts staff, school administrators, students, and parents, with permissions appropriate to each role.

| Domain | Proposed digital workflow | Quality control focus |
|---|---|---|
| Attendance | Digital roll call and attendance records | Accuracy, retrieval speed, audit trail |
| Fees | Fee ledger, payment status, receipts | Validation, reconciliation, traceability |
| Academics | Marks, gradebook, report cards | Completeness, controlled access, fewer compilation errors |
| Timetable | Timetable creation and access | Conflict checks and clear communication |
| Promotion | Promotion eligibility review | Consistent rules and documented decisions |
| PTM / Leave | Parent–teacher meeting and leave requests | Faster approvals, status visibility |

# 3. TQM Principles Applied

**Quality as performance versus expectations (Q = P/E).** The guide uses Q = P/E to frame quality as the relationship between actual performance (P) and stakeholder expectations (E). The project seeks to raise actual

service performance through faster workflows, fewer errors, and better communication.

**Deming's Chain Reaction.** Improve quality → decrease costs → improve productivity → delight stakeholders → support institutional sustainability. Digital workflows are intended to reduce rework and improve service consistency.

**Deming's 14 Points.** The project aligns with management commitment, continuous training, reducing fear around reporting errors, removing functional silos, and sustaining improvement rather than relying on slogans alone.

**Juran's Quality Trilogy.** Quality planning is reflected in system architecture and workflow design; quality control in validation rules and audit checks; quality improvement in recurring Kaizen cycles based on user feedback and defect trends.

**Crosby's Zero Defects.** Preventive validation can block invalid transactions, such as negative payments, overpayments, room double-bookings, and PTM schedule clashes, where these controls are implemented.

# 4. Customer Focus and Satisfaction

**Internal customers:** teachers, accountants, bus drivers, and the principal. **External customers:** students, parents, and education regulators. Designing around both groups helps ensure that process improvements are useful across the institution.

**Kano model.**
- **Must-be requirements:** accurate fee balances, complete marks, and protected student data.
- **One-dimensional satisfiers:** online fee payments, instant report-card downloads, and online leave requests.
- **Attractive delighters:** live bus tracking with ETA alerts, automated teacher substitution, and promotion eligibility audits.

**SERVQUAL.**
- **Tangibles:** a clear, usable web interface instead of scattered paper registers.
- **Reliability:** accurate ledgers and dependable student records.
- **Responsiveness:** prompt receipts and leave approvals.
- **Assurance:** authentication and role-based access control.
- **Empathy:** multilingual support and appropriate handling of approved medical leave.

The guide also states that retaining an existing student costs five times less than acquiring a new one and attributes 68% of customer defection to employee attitude and bureaucratic friction. These are reproduced as guide claims and should be validated before external use.

# 5. The Seven Quality Control Tools

The seven QC tools create a practical route from defect observation to root-cause analysis and ongoing control. The specific counts and statistics below are those supplied in the project guide.

## 1. Check Sheet

A 60-day audit across 50 schools is described as recording 741 defect incidents. A check sheet provides a consistent structure for collecting defect type, frequency, date, and department.

## 2. Stratification

The guide divides defects by locality (rural 57.4%, urban 42.6%) and department (accounts 42.1%, academics 40.3%, logistics 17.6%). Stratification helps identify where improvement work should begin.

## 3. Pareto Chart

The top three defect categories are fee calculation mismatches (42.1%), attendance register losses (24.8%), and gradebook compilation errors (15.5%). Together, these account for 82.4% in the guide, illustrating the Pareto principle.

## 4. Cause-and-Effect / Fishbone

A 6M analysis groups possible root causes into Manpower, Method, Machine, Material, Measurement, and Milieu (environment). It supports investigation rather than assuming that individual staff are the sole cause.

### 5. Histogram

The guide reports turnaround time shifting from a broad, right-skewed distribution with mean 48.2 hours to a centered distribution with mean 1.8 hours. A histogram can show spread, skew, and whether service times cluster around a target.

### 6. p-Chart

The guide states p-bar = 0.125, UCL = 0.224, LCL = 0.026 for 15 daily lots of 100 transactions. Day 5 is reported at p = 0.28, above the UCL, associated with month-end fee rush. Post-implementation defect rate is stated as below 0.8%.

### 7. Scatter Diagram

The guide reports Pearson r = +0.94 and $R^2$ = 0.884 between paperwork hours and operational errors. This indicates a strong positive association in the described data; correlation alone does not prove causation.

## 6. 5S Lean Workplace and Waste Reduction

**Seiri — Sort:** identify and remove obsolete registers and vouchers according to retention rules.
**Seiton — Set in Order:** organize digital records using a relational database, consistent identifiers, and indexed search.
**Seiso — Shine:** maintain clean records, monitor logs, and schedule data-quality checks.
**Seiketsu — Standardize:** apply consistent validation rules and standard operating procedures across workflows.
**Shitsuke — Sustain:** hold monthly quality-circle meetings, train staff, and review improvement actions.

The guide also frames waste through the 3Ms: Muri (overburden), Mura (unevenness), and Muda (non-value-adding work). Digital workflows aim to reduce unnecessary movement of files, waiting in queues, paper inventory, duplicate entry, and rework. Digitalization still requires disciplined backups, access controls, retention policies, and staff training.

## 7. Process Capability and Cost of Quality

The supplied guide specifies service turnaround-time limits of LSL = 0 hours and USL = 24 hours. It reports pre-TQM capability of Cp = 0.27 and Cpk = -0.55, compared with post-TQM Cp = 6.45 and Cpk = 1.82. These values are reproduced from the guide; a formal capability claim should be supported by the underlying observations, stable-process evidence, and calculation method.

**Cost of Quality (COQ).** The guide estimates annual cost at ■52,00,000 before implementation and ■13,60,000 after implementation, a stated net saving of ■38,40,000 (73.8%) and payback under three months. These are scenario estimates from the supplied guide, not independently audited financial results.

## 8. Quantitative Impact Scorecard

The table below reproduces the scorecard in the supplied project guide. Treat the right-hand figures as proposed targets/benchmarks unless validated with pilot measurements and a documented sample.

| Metric | Paper baseline | Digital TQM benchmark | Stated gain |
|---|---|---|---|
| Roll call attendance time | 15–20 min/class | 30 sec | 96.7% time saved |
| Fee ledger reconciliation | 14 days/monthly audit | Real-time | 100% automated |
| Report-card lead time | 21 days post-exams | One-click | 95.2% reduction |
| Annual paper consumption | 45,000 sheets/school | 0 sheets | 100% eliminated |

| Metric | Paper baseline | Digital TQM benchmark | Stated gain |
|---|---|---|---|
| Student record loss | 14.2% historical loss | 0.00% stated | Zero-defect target |
| Parent satisfaction (CSAT) | 38.4% | 94.6% | +146.3% stated |
| Process capability (Cpk) | -0.55 | 1.82 | Process capable |
| Annual quality cost | ■52,00,000 | ■13,60,000 | 73.8% reduction |

# 9. Implementation Roadmap

**Phase 1 — Discover:** map existing paper workflows, identify stakeholders, and record baseline cycle times and error rates.
**Phase 2 — Design:** define roles, permissions, data fields, validation rules, backup procedures, and retention policy.
**Phase 3 — Pilot:** introduce selected workflows in one department or grade; train users and capture issues.
**Phase 4 — Measure:** compare baseline and post-pilot measures using check sheets, Pareto analysis, and control charts where suitable.
**Phase 5 — Improve:** prioritize defects, assign owners, run Kaizen cycles, and update standard operating procedures.
**Phase 6 — Scale:** expand gradually after usability, security, reliability, and support processes are demonstrated.

# 10. Risks and Mitigations

- **Data privacy:** use least-privilege access, secure authentication, backups, and clear retention rules.
- **Adoption resistance:** provide role-specific training, feedback channels, and a gradual transition plan.
- **Connectivity constraints:** test performance on low-bandwidth connections and plan for downtime procedures.
- **Data migration errors:** reconcile migrated records and maintain an auditable correction process.
- **Overclaiming impact:** distinguish target values from measured results and publish only validated outcomes.

# 11. Conclusion

The School Management System is a practical setting for applying TQM to a service institution. Its value is not merely replacing paper with screens: it is standardizing work, preventing errors, improving access to information, listening to internal and external customers, and using data to continuously improve processes. The strongest submission will pair the proposed system with a small, documented pilot and transparent before-and-after measurements.

# Presentation delivery tips

Open with the real-world problem of manual paperwork. Connect the project to Q = P/E, Deming's chain reaction, Kano, SERVQUAL, and 5S. Then explain the seven QC tools in sequence: check sheet → stratification → Pareto → fishbone → histogram → p-chart → scatter diagram. Clearly label the guide's numerical values as baseline/benchmark figures unless your team has evidence to verify them. If demonstrating the prototype, confirm that it is running locally before presenting the localhost address.

## Reference note

Primary source for this generated report: user-provided file, TQM_Project_Presentation_Guide.md. No external sources were added. All project-specific statistics were carried over from that guide and have not been independently verified.